

Titanic: class X and survival Y

died ($Y = 0$) survived ($Y = 1$) $P(X = x)$

$X = 1$	0.090	0.153	0.242
$X = 2$	0.109	0.098	0.207
$X = 3$	0.418	0.134	0.551
$P(Y = y)$	0.616	0.384	1

joint (inner cells):
 $P(X = x \cap Y = y)$, sum to 1

marginal (the margins):
row and column sums

conditional:
 $P(Y = 0 \mid X = 3) = 0.418/0.551 = 0.758$